

BUBX: Technical Description

Airflow Demand Calculator for Submerged Air Diffusion Systems

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Purpose and Overview

BUBX is a web-based application for calculating the airflow demands and analyzing the hydraulic performance of submerged air diffusion manifolds, such as the air screens and bubbler systems used for ice control at navigation projects. It is a Python reimplement and extension of the legacy CRREL Fortran program BUB300. Given a system geometry (supply line, manifold, and orifices) and a set of boundary conditions, BUBX computes the discharge from every orifice, the pressure distribution along the manifold, the total air demand, and the expected induced surface velocity. This supplement summarizes the governing equations and solution method implemented in the code; broader design guidance for air systems is provided in the companion memorandum, “Design of Air Systems for Ice Control” (Engel, 2021).

The user interacts with BUBX through a browser interface (built on the Streamlit framework) organized into tabs for inputs, tabulated results, orifice flow plots, and flow-range (sensitivity) analysis. Results can be exported as a PDF report that records the inputs and outputs of a run for project documentation.

Model Structure

BUBX represents an air system as an ordered sequence of two kinds of features: pipe segments and orifices. A typical model consists of a supply line, followed by alternating manifold pipe segments and orifices, ending at the far end of the diffuser. Each pipe segment carries a length, diameter, and absolute roughness (selectable by material); each orifice carries a diameter, a contraction coefficient, and an elevation. Orifice elevations may be assigned from a user-specified elevation profile along the manifold, so systems installed on sloping or irregular bathymetry are represented with the correct local hydrostatic pressure at each orifice.

Governing Equations

Orifice discharge and choked flow

Air discharge through each orifice is computed with the compressible mass flow rate equations of Jobson (1955). The general equation for the mass flow rate through an orifice is

$$\dot{m} = C K_N A \sqrt{P_0 \rho_0} \quad (1)$$

where C = the orifice contraction coefficient; K_N = a mass flow rate coefficient which varies with the flow condition; A = the projected area of the orifice; P_0 = the absolute pressure upstream of the orifice (inside the manifold); and ρ_0 = the density of the air on the upstream side of the orifice, computed from the ideal gas law at the local pressure and the specified air temperature. For subsonic (subcritical) flow,

$$K_N = \sqrt{\frac{2\gamma}{\gamma-1} r^{\frac{2}{\gamma}} (1 - r^{\frac{\gamma-1}{\gamma}})} \quad (2)$$

where γ = the ratio of specific heats (1.4 for air) and r = the ratio of downstream to upstream absolute pressure across the orifice. The downstream pressure at each orifice is the local hydrostatic pressure, determined from the water depth and the orifice elevation. When r falls below the critical pressure ratio

$$r_c = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} = 0.5283 \quad (3)$$

the flow through the orifice reaches the local speed of sound and becomes “choked”: the mass flow rate is controlled only by the upstream pressure and the orifice geometry, and is no longer influenced by the downstream water pressure. In this supercritical condition the coefficient takes the constant value

$$K_N = \sqrt{\gamma \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}} \quad (4)$$

BUBX evaluates the pressure ratio and applies the appropriate coefficient, Equation (2) or (4), independently at every orifice on every iteration, so a single system can contain both choked and unchoked orifices — for example where the manifold spans a range of depths. Experiments by Leibson et al. (1956) on air discharged through submerged orifices into water showed behavior consistent with the air-air theory provided the orifice throat contains the *vena contracta*, with measured discharge coefficients near 0.9 as flow approached choked conditions. Reflecting those results, the default orifice contraction coefficient in BUBX is $C = 0.9$; the value can be adjusted by the user.

Pipe friction losses

Pressure drop in the supply line and in each manifold segment is computed with the Darcy-Weisbach equation:

$$\Delta P = f \frac{L}{D} \frac{\rho v^2}{2} \quad (5)$$

L = the length of the pipe segment; D = the pipe diameter; ρ = the density of the air; v = the velocity of the air in the pipe; and f = the Darcy friction factor. For the turbulent flows typical of air systems (Reynolds number $> 4,000$), f is obtained from the implicit Colebrook relation (Moody, 1944):

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re \sqrt{f}} \right) \quad (6)$$

where ε = the absolute pipe roughness. BUBX solves Equation (6) using Clamond's rapid method as implemented in the open-source “fluids” library, which was verified against the iterative friction factor routine in legacy BUB300 with negligible differences. Because the air is compressible, its density (and therefore velocity) changes along a segment as pressure drops. Following the guidance in Crane Technical Paper 410, BUBX evaluates Equation (5) using the average pressure and density in each segment, solved iteratively; this refinement matters mainly for long, high-flow segments and reduces to the simpler upstream-pressure treatment for the short segments typical between orifices. The Reynolds number and Mach number of each segment are reported so the user can confirm the flow regime remains within the validity of the method.

Induced surface velocity

For air screens, the horizontal surface current generated by the rising bubble plume is the primary performance measure. BUBX estimates the near-surface water velocity using the empirical power-law

relationship of Haehnel (2016), which modified the fit of Tuthill and Stockstill (2005) to include the weak dependence on submergence depth observed in field and laboratory data:

$$V = \frac{351 + 25.47H}{10} Q_a^{0.4223} \quad (7)$$

where V = the average near-surface water velocity (cm/s); H = the depth of manifold submergence (m); and Q_a = the airflow per unit length of manifold (SCMM/m). BUBX reports the surface velocity both as a system average and locally at each orifice, using the local airflow per unit length, so that variation in holdback capability along the screen is visible.

Flow uniformity

Non-uniform discharge along a manifold (strong near the compressor, weak at the far end) can leave gaps in an air screen even when the total airflow is adequate. BUBX quantifies uniformity with a coefficient based on Haehnel (2016), rescaled so that a value of 1.0 indicates perfectly uniform flow:

$$U_C = 1 - \frac{Q_{max} - Q_{min}}{Q_{avg}} \quad (8)$$

where Q_{max} , Q_{min} , and Q_{avg} are the maximum, minimum, and average orifice discharges. The interface flags designs with U_C below 0.9; new designs should target values near 1.0.

Solution Procedure

The equations above are coupled: the discharge at each orifice depends on the local manifold pressure, which depends on the friction losses upstream, which depend on the flow remaining in the line, which depends on the discharge at every upstream orifice. BUBX resolves this with a downstream-marching scheme wrapped in a bracketed bisection solver which is a robust replacement for the fixed-relaxation update used in legacy BUB300, which could fail to converge for operating points near hydrostatic pressure. The primary mode of operation specifies the compressor (upstream) pressure and solves for the total airflow:

1. A trial total mass flow rate is selected from within a bracketing interval.
2. The solution marches downstream feature by feature: the pressure drop across each pipe segment is computed with Equations (5) and (6) from the mass flow remaining in the line, and the discharge through each orifice is computed with Equations (1)–(4) from the local internal pressure and subtracted from the remaining mass flow.
3. The residual mass flow at the end of the line indicates whether the trial was too high or too low (guard conditions also flag physically invalid states, such as internal pressure falling below local hydrostatic pressure).
4. The bisection interval is narrowed and the march repeated until the residual falls below tolerance. The converged solution gives the discharge at every orifice and the pressure at every point in the system for that supply pressure.

The solver can be driven by any one of three design criteria:

Pressure specified. The supply pressure is given directly and the procedure above returns the resulting total airflow and orifice discharge distribution.

Airflow specified. The user specifies a target total airflow and BUBX performs an outer bisection on supply pressure, running the downstream solver at each trial pressure, until the computed system airflow matches the target. This directly answers the common design question of what pressure is required to deliver a specified air demand.

Surface velocity specified. The user specifies a target induced surface velocity, which is converted to the required airflow per unit length of manifold by inverting the Haehnel relationship, Equation (7), at the operating depth. The corresponding total airflow (unit airflow times diffuser length) is then solved for supply pressure as in the airflow-specified mode. The target is met as a system average; local velocities along the screen will vary with the orifice flow distribution and are reported in the results. This allows a screen to be sized directly from its performance requirement; the current needed to hold back or deflect ice.

Inputs and Outputs

Inputs: manifold segment length; number of orifices or orifice spacing; orifice diameter; manifold and supply pipe diameters, lengths, and materials (roughness); an optional elevation profile and datum for the diffuser; water depth; air temperature; and one of three boundary conditions: the supply pressure, the target total airflow, or the target surface velocity. Units may be toggled between English and SI.

Outputs: total system airflow; required or specified supply pressure; airflow per unit length of manifold; the flow distribution across all orifices with choked/unchoked status; pipe velocities, pressure drops, Reynolds and Mach numbers; the predicted surface velocity, Equation (7); and the uniformity coefficient, Equation (8). Standard flow rates are reported at the 20°C (68°F) reference temperature used by North American compressor suppliers, with the 0°C convention used historically for ice-control systems also noted, an airflow specified at the 0°C reference is approximately 7% larger when expressed at the 20°C reference.

Availability

BUBX is open source and available at <https://github.com/ski907/BUBX>. The core calculation engine (solver, geometry, and physics modules) is separable from the web interface and can be scripted directly for batch studies or custom configurations.

References

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